

SAATHI

One Financial Memory — Bank, Cash, SHG, Chits, Post Office & Business, Together

Final Build Plan — Version 3 (Post Council Review)

Single source of truth for Person A, B, C, D and any build agent

This version supersedes all earlier drafts. It folds in: (1) the Twilio WhatsApp Sandbox + load-rebalance decision, and (2) the five-agent council review — Contrarian, First Principles, Expansionist, Outsider, Executor. The council's core verdict: the idea is strong, the scope was too wide, the innovation (Money Pot Map) wasn't visible enough, and integration depended on unfinished components. This draft cuts scope, makes contracts mock-first, and centers the whole build and the whole pitch on ONE woman's ONE financial story.

1. The 10-Second Pitch (say this first, always)

A rural woman's money is never in one place. Some is in her hand, some in a bank, some in her SHG, some locked in a chit fund, some in the Post Office, and some tied up in a small business she runs. Nobody — including her — sees all of it at once. Saathi brings it into one picture, in her own language, over WhatsApp, and helps her make her next financial decision.

Outsider (council): the document was written for engineers before it explained what Saathi does for a woman. Every teammate opens the pitch with the paragraph above — architecture comes after, never before.

The one-line internal compass

If a feature doesn't help show “her money, in one place, leading to one better decision” inside a 3-minute demo, it is roadmap, not build. This sentence overrides every other section of this document if they conflict.

2. The Demo Story (build and pitch both revolve around this)

Per the council: the demo is not a tour of ten features. It is one woman, one continuous story, told once, end-to-end.

Meera's story (demo persona — use this exact seed data everywhere)

- She does tailoring work and sells pickles on the side.
- On WhatsApp voice note: “I earned ₹800 from tailoring today.” → Saathi logs it as business income.
- Her money currently sits across: Cash ₹2,000 · Bank ₹5,000 · SHG ₹2,500 · Chit committed ₹4,000 · Post Office ₹5,000.
- She has a goal: ₹20,000 for her daughter's education, ₹8,000 saved so far.
- Saathi shows her the whole picture in one view (Money Pot Map), flags that her chit commitment is eating into what she could save monthly, and suggests — at category level, never a named product — a way to close the education-goal gap faster.
- A suspicious SMS arrives (“Your KYC will expire, click to verify”) — Safety Shield flags it in plain language.

This exact JSON is the shared demo fixture — every teammate builds and tests against it so the four workstreams snap together without last-minute data mismatches: { "cash": 2000, "bank": 5000, "shg": 2500, "chit_committed": 4000, "post_office": 5000, "business": { "activity": "pickle sales + tailoring", "last_entry": { "revenue": 1000, "cost": 600, "profit": 400 } }, "goal": { "name": "Education", "target": 20000, "saved": 8000 } }

3. Scope — What We Build, What We Cut

The Contrarian's core warning: this architecture (WhatsApp + voice + SMS + RAG + literacy tiers + simulator + safety shield + business ledger + marketplace + dashboard + React Native) is too much for 24–36 hours, and it chains A → B → C → D so one failure blocks everyone. This section is the binding cut list — if it's not in “Must-have,” it does not get built before the demo is safe.

Tier	In scope	Why
Must-have	Twilio WhatsApp Sandbox text+voice input · Money Pot Map (visual, live) · RAG education + personalized guidance · Literacy-tier quiz · One Decision Simulator scenario · Safety Shield (SMS scam check) · Business ledger (single entry → revenue/cost/profit)	This alone tells Meera's whole story end to end
Should-have	React Native app shell mirroring the WhatsApp flow · OCR for receipts · Basic marketplace listing generated from one ledger entry	Strengthens the “and she earns more” moment if time allows

Tier	In scope	Why
Roadmap only (pitch slide, not built)	IVR / missed-call voice · Full SMS auto-capture / inbox parsing · Multi-scenario simulator · Full 12-module curriculum · SHG/lender partner integrations	Highest effort, lowest demo value, or depends on access Claude cannot verify you'll have (see Section 3.1)

3.1 SMS auto-capture — verify before you rely on it

Contrarian flag: the plan assumed access to a rural user's SMS inbox for auto-parsing. On Android this needs the READ_SMS/RECEIVE_SMS runtime permission granted inside your own app (not achievable from a WhatsApp/Twilio number, which never sees the user's native SMS inbox). Before this is treated as a demo feature: confirm your test phone can grant that permission to a companion app you control. If not confirmed by Hour 6, SMS input is simulated with a canned demo message and the feature stays on the roadmap slide only. Voice note and typed WhatsApp messages are not affected — they remain must-have.

4. Make the Innovation Visible (First Principles feedback)

The Money Pot Map is the actual innovation, but as a backend concept (Postgres tables + an API) it is invisible to a judge who only sees a WhatsApp chat. Person D's dashboard has one job above all others: make the unified picture obvious within seconds, with no explanation needed.

What the dashboard must show, literally

- A single total: “₹19,500 across 5 pots” rendered large, at the top.
- Each pot as its own visual block with amount: Bank → ₹5,000 · Cash → ₹2,000 · SHG → ₹2,500 · Chit (committed) → ₹4,000 · Post Office → ₹5,000.
- The education goal shown as progress toward ₹20,000, not buried in a table.
- No architecture, no database language, no jargon anywhere on this screen — a judge should understand it before you finish a sentence.

First Principles' framing to keep repeating to the team: “Don't demo the architecture. Demo the new mental model.”

5. Protect the Income-Generation Story (Expansionist feedback)

The Expansionist's view: the project currently reads as “we help her manage money.” The Informal Business Ledger lets it also say “we help her earn more” — the “and it pays for itself” moment. Even under time pressure, this stays in must-have, at minimum as a single flow:

1. She reports one sale by voice (“I sold 10 pickle bottles for ₹1,000”).
2. Saathi extracts revenue (₹1,000), asks or infers cost (₹600), computes profit (₹400).
3. That profit entry updates her Money Pot Map (business pot) and, if time allows, offers to turn it into a one-tap marketplace listing.

If only one Should-have item survives triage, this is the one the Expansionist would keep — it converts financial tracking into a growth narrative, not just a bookkeeping one.

6. One Simple Message (replaces the 7-step loop)

The Outsider flagged the original loop — TRACK → UNDERSTAND → LEARN → DECIDE → ACT → MEASURE → PERSONALIZE — as too many words for a 3-minute demo to land. It's still fine as an internal design reference, but the team, the slides, and the demo narration should use exactly this three-word version instead:

SEE. DECIDE. ACT.

- SEE — Money Pot Map shows her whole financial picture in one view.
- DECIDE — RAG-grounded guidance + Decision Simulator help her weigh one real choice.
- ACT — she logs income, avoids a scam, or moves toward her goal — visibly, on screen.

7. Contracts Before Implementation (Executor feedback)

The Executor's finding: both Person C and Person D depend on Person B's financial-state API, but that API wasn't ready until roughly Hour 16 — meaning two people would sit blocked on one. Fix: every cross-team dependency ships a mock at Hour 0–2, and the real implementation swaps in later without changing anyone else's code.

Owner	Contract they publish by Hour 2	Consumed by
Person B	Mock financial-state JSON (see Section 2 fixture) served from a stub endpoint or a static file	Person C, Person D
Person A	Mock normalized-input JSON: user_id, channel, input_type, raw_text, normalized_text, parsed_transaction {type, amount, category}, confidence	Person B, Person C
Person C	Mock response JSON: response_text, source_class (RBI/SEBI/NCFE vs research vs provider), mode (education/personalized), disclaimer flag	Person A, Person D

Rule: nobody waits on another person's real backend before Hour 16. Everyone builds and demos against these mocks first, then does a single integration pass once real endpoints exist. This is the direct fix for the $A \rightarrow B \rightarrow C \rightarrow D$ blocking chain the Contrarian and Executor both raised.

8. System Architecture

Person A (WhatsApp/voice input) normalizes what the user says → Person B stores and serves her unified financial state (the Money Pot Map) → Person C reads B's state, decides what to teach or say, grounded in retrieval, and returns a response → Person A delivers that response back to her → Person D's dashboard reads B's state directly to render the visible Money Pot Map and the business ledger/marketplace view, and calls Person C for any AI-generated marketplace text.

- Person C is the only component allowed to touch the LLM for financial content — this is the compliance boundary and does not change in this revision.
- All four workstreams build against the Section 7 mocks first; real integration is one deliberate pass, not a continuous dependency.

9. Team Split — Four Workstreams

Person	Owns	Depends on
A	Twilio WhatsApp Sandbox bot (text + voice) · Whisper STT · normalized-input contract · SMS parsing/fraud flagging (only if 3.1 is confirmed feasible — else canned demo input)	Nothing blocking — builds standalone, publishes mock at Hour 0–2
B	Money Pot Map data model · financial-state API (mock → real) · business ledger storage · duplicate detection · goals · audit log	Nothing blocking — publishes mock at Hour 0–2
C	RAG pipeline · education + personalized guidance modes · literacy-tier quiz · one Decision Simulator scenario · Safety Shield · escalation logic	B's mock (Hour 0–2), B's real API by integration pass
D	Money Pot Map dashboard (must be judge-legible in seconds, Section 4) · business ledger view · marketplace (should-have) · React Native app shell (should-have) · OCR (should-have) · demo script/slides	B's mock for dashboard data, C's response contract for AI text

Unchanged from the prior revision: Twilio WhatsApp Sandbox (not the Meta Business API) for fast, verification-free setup; React Native shell and OCR moved from A to D to rebalance load; IVR/missed-call voice dropped from build scope entirely and kept as a pitch-only roadmap line.

10. Hour-by-Hour Build Order (all four, in parallel)

Hour 0–2 — Contracts first, before any real logic

- Every person publishes their mock (Section 7). Nothing else starts until this exists — it is what makes the rest of the day parallel instead of sequential.

Hour 2–6

- A: Sandbox joined, webhook receiving messages, Whisper wired for voice notes.
- B: Real data model stood up behind the same mock contract shape.
- C: Vector store (Chroma) stood up, 2–3 curated modules ingested (Money Map, Saving & Goals, Loans).
- D: Dashboard skeleton rendering the mock fixture as static pots.

Hour 6–12

- A: Voice → text → normalized-input contract working end to end against a live sandbox number.
- B: Real financial-state API live; duplicate detection and goal storage working.
- C: RAG + education mode working standalone against mocked state — does not wait for B's real API.

- D: Dashboard wired to B's mock live-updating; business ledger view started.

Hour 12–18

- A ↔ B integration: real normalized input flowing into real financial-state storage.
- C: personalized-guidance mode wired to B's real API; literacy-tier quiz live and changes wording.
- D: Money Pot Map fully live off real data; marketplace/OCR/React Native shell if time allows.

Hour 18–24

- C: Decision Simulator for the one chosen scenario (loan comparison or savings-goal feasibility).
- Full A→B→C→D integration pass — the one deliberate cutover from mocks to real endpoints.
- Run Meera's story start to finish at least twice, on the actual demo device.

Hour 24+

- Safety Shield + escalation logic if time remains.
- Demo script locked to the SEE → DECIDE → ACT narrative (Section 6); rehearse against Meera's fixture only — no live improvisation with new numbers on demo day.

11. Tech Stack

Layer	Stack
Channel (A)	Twilio WhatsApp Sandbox · Whisper (STT) · webhook service (Node/Express or FastAPI)
Data / Money Pot Map (B)	Postgres · REST API · JSON contract as in Section 2/7
Brain (C)	Chroma (vector store) · LLM via API · prompt-conditioned literacy tiers · rule-based Safety Shield pattern checks
Interface (D)	Web dashboard (React) · React Native app shell (should-have) · OCR library (should-have)

12. Testing Plan

- Unit level: each mock contract validated independently before any cross-team call is made (A's normalized JSON, B's financial-state JSON, C's response JSON).
- Integration: run Meera's exact fixture through the full A→B→C→D chain at Hour 18–24 and again 1–2 hours before the demo slot — no new data introduced between those two runs.
- RAG grounding check: every education/personalized response manually checked against its source_class tag — confirm nothing is answered outside the 2–3 seeded modules.
- Safety Shield: test against 3–4 canned suspicious SMS examples plus 1–2 legitimate ones, confirm it never states certainty (“looks suspicious because X,” never “this is a scam”).
- Literacy tiers: same question run at Tier 1 and Tier 3, confirm visibly different vocabulary/length — this is a demoable check, not just a backend one.
- Failure mode: if any Should-have/Roadmap item is unstable 2 hours before demo, cut it from the live run and fall back to the slide version — must-have items are the only ones allowed on stage.

13. Demo Script (3 minutes, Meera end to end)

4. 0:00–0:20 — One-line pitch (Section 1), no architecture talk.

5. 0:20–0:50 — SEE: Meera sends a WhatsApp voice note (“I earned ₹800 from tailoring”); Money Pot Map appears live on the dashboard, whole picture visible in seconds.
6. 0:50–1:30 — DECIDE: ask Saathi something about her education goal; show Tier 1 vs Tier 3 answer briefly; show the Decision Simulator scenario.
7. 1:30–2:00 — ACT: log the pickle-sale income → profit computed → pot updates → (if built) one-tap marketplace listing.
8. 2:00–2:20 — Safety Shield: paste the suspicious KYC SMS, show the plain-language flag.
9. 2:20–3:00 — Close on the SEE → DECIDE → ACT line and the roadmap slide (IVR, full SMS capture, full curriculum) — named explicitly as “what’s next,” not apologized for.

14. Open Decisions — Resolve Once, Up Front

- Confirm SMS auto-capture feasibility per Section 3.1 by Hour 6, or lock in the canned-input fallback.
- Pick the single Decision Simulator scenario — loan comparison or savings-goal feasibility — before Hour 18.
- Confirm the exact 2–3 curriculum modules seeded into RAG (recommended: Money Map, Saving & Goals, Loans).
- Decide, by Hour 12, whether React Native shell / OCR / marketplace (Should-have) are still realistic given progress — if not, drop cleanly rather than half-build.
- Confirm all four workstream owners against this document — owners are assigned by function (A/B/C/D), not by who wrote which research doc originally.

This document is the single source of truth for the build. If any teammate's verbal understanding conflicts with what's written here, this document wins — update it in place rather than working from a private version.